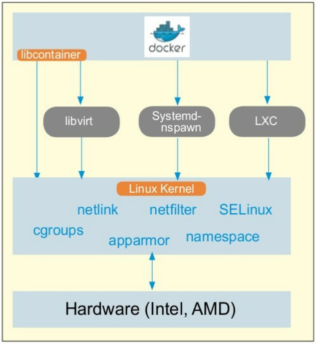

Figure 1: Linux Container

Containerisation vs virtualisation

What is the rationale behind the container-based approach or how is it different from virtualisation? Figure 2 speaks for itself.

Containers virtualise at the OS level, whereas both Type-I and Type-2 hypervisor-based solutions virtualise at the hardware level. Both virtualisation and containerisation are a kind of virtualisation; in the case of VMs, a hypervisor (both for Type-I and Type-II) slices the hardware, but containers make available protected portions of the OS. They effectively virtualise the OS. If you run multiple containers on the same host, no container will come to know that it is sharing the same resources because each container has its own abstraction.LXC takes the help of name spaces to provide the isolated regions known as containers. Each container runs in its own allocated name space and does not have access outside of it. Technologies such as cgroups, union filesystems and container formats are also used for different purposes throughout the containerisation.

Linux containers

Unlike virtual machines, with the help of LXC you can share multiple containers from a single source disk OS image. LXC is very lightweight, has a faster start-up and needs less resources.

Installation of Docker

Before we jump into the installation process, we should be aware of certain terms commonly used in Docker documentation.

Image: An image is a read-only layer used to build a container.

Container: This is a self-contained runtime environment that is built using one or more images. It also allows us to commit changes to a container and create an image.

Docker registry: These are the public or private servers, where anyone can upload their repositories so that they can be easily shared.

The detailed architecture is outside the scope of this article. Have a look at *http://docker.io* for detailed information.

Note: I am using CentOS, so the following instructions are applicable for CentOS 6.5.

Docker is part of Extra Packages for Enterprise Linux (EPEL), which is a community repository of non-standard packages for the RHEL distribution. First, we need to install the EPEL repository using the command shown below:

```
[root@localhost ~] # rpm -ivh http://dl.fedoraproject.org/
pub/epel/6/x86_64/epel-release-6-8.noarch.rpm
```

As per the best practice update,

```
[root@localhost ~] # yum update -y
```

docker-io is the package that we need to install. As I am using CentOS, Yum is my package manager; so depending on your distribution ensure that the correct command is used, as shown below:

```
[root@localhost ~] # yum -y install docker-io
```

Once the above installation is done, start the Docker service with the help of the command below:

```
[root@localhost ~] # service docker start
```

To ensure that the Docker service starts at each reboot, use the following command:

```
[root@localhost ~] # chkconfig docker on
```

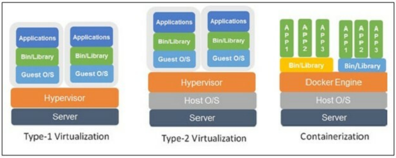

Figure 2: Virtualisation